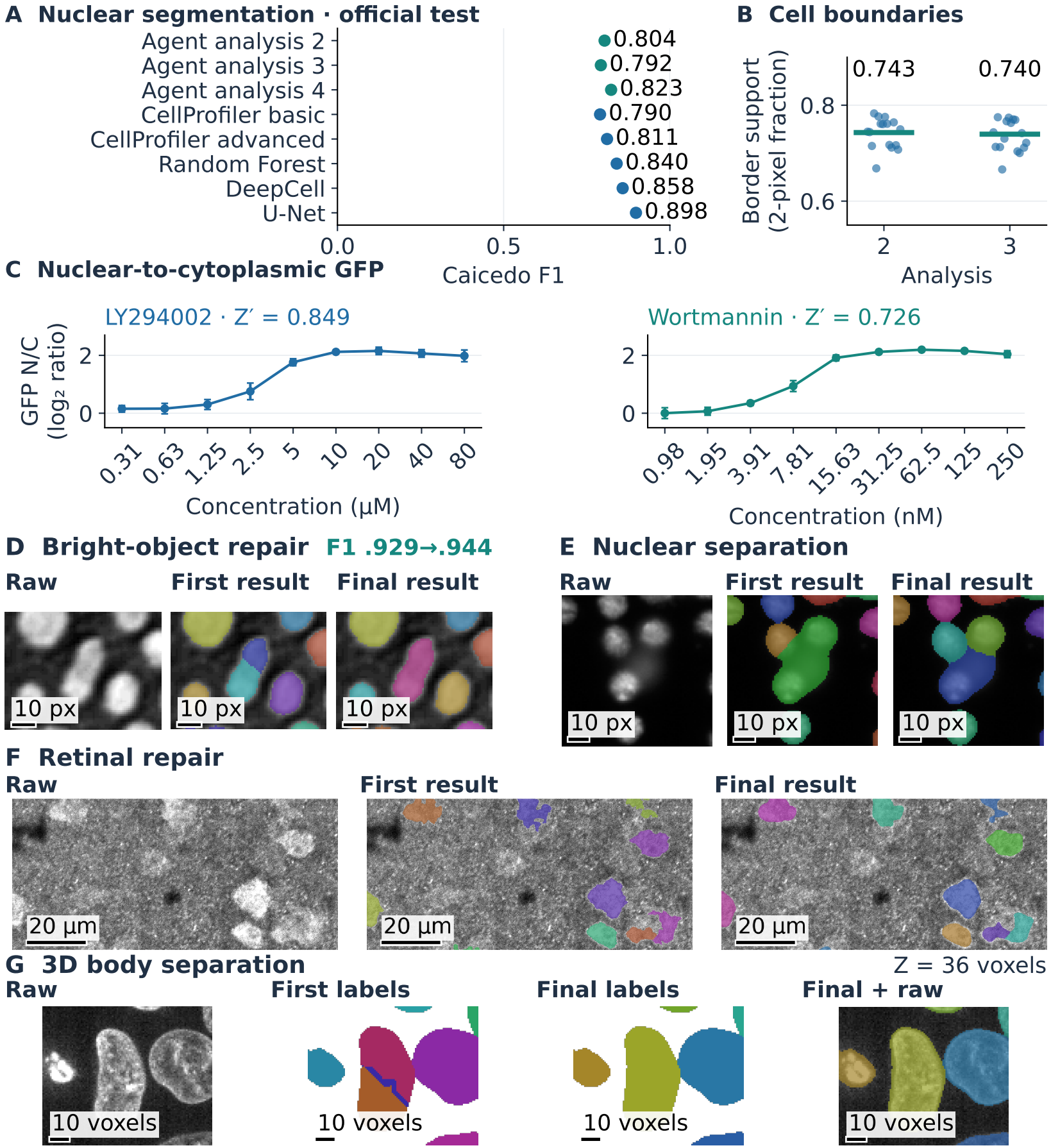

A Nuclear segmentation · official test
Agent analysis 2
Agent analysis 3
Agent analysis 4
CellProfiler basic
CellProfiler advanced
Random Forest
DeepCell
U-Net
0.804
0.792
0.823
0.790
0.811
0.840
0.858
0.898
0.0
0.5
1.0
Caicedo F1
B Cell boundaries
Border support (2-pixel fraction)
0.743
0.740
0.8
0.6
2
3
Analysis
C Nuclear-to-cytoplasmic GFP
LY294002 · Z′ = 0.849
GFP N/C (log₂ ratio)
2
0
0.31
0.63
1.25
2.5
5
10
20
40
80
Concentration (μM)
Wortmannin · Z′ = 0.726
2
0
0.98
1.95
3.91
7.81
15.63
31.25
62.5
125
250
Concentration (nM)
D Bright-object repair F1 .929→.944
Raw
First result
Final result
10 px
10 px
10 px
E Nuclear separation
Raw
First result
Final result
10 px
10 px
10 px
F Retinal repair
Raw
First result
Final result
20 μm
20 μm
20 μm
G 3D body separation
Z = 36 voxels
Raw
First labels
Final labels
Final + raw
10 voxels
10 voxels
10 voxels
10 voxels